Package 'Satisficing'

October 18, 2026

Type Package

Title Tools for Detecting Indicators of Survey Satisficing

Version 0.4.0

Description Provides tools for detecting potential indicators of satisficing in survey responses, including straight-line response patterns and row-wise evaluation of logical conditions, as well as careless and insufficient effort response indices ported from the 'careless' package (including longstring, intra-individual response variability, even-odd consistency, psychometric synonyms/antonyms, and Mahalanobis distance).

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URL https://github.com/soheidon/Satisficing

BugReports https://github.com/soheidon/Satisficing/issues

Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Imports psych

Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

Config/roxygen2/version 8.1.1

RoxygenNote 7.3.3

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`careless_dataset` *Simulated dataset with insufficient effort responses.*

Description

A simulated dataset mimicking insufficient effort responding. Contains three types of responses: (a) Normal responses with answers centering around a trait/attitude value (80 percent probability per simulated observation), (b) Straightlining responses (10 percent probability per simulated observation), (c) Random responses (10 percent probability per simulated observation). Simulated are 10 subscales of 5 items each (= 50 variables). Ported from the **careless** package (version 1.2.2).

Usage

```
careless_dataset
```

Format

A data frame with 200 observations (rows) and 50 variables (columns).

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

careless_dataset2 *Simulated dataset with careless responses.*

Description

A simulated dataset mimicking insufficient effort responding. Contains three types of responses: (a) Normal responses with answers mimicking a diligent respondent (b) Some number of longstring careless responders, (c) some number of generally careless responders. Simulated are 10 subscales of 10 items each (= 100 variables). Ported from the **careless** package (version 1.2.2).

Usage

```
careless_dataset2
```

Format

A data frame with 1000 observations (rows) and 100 variables (columns).

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

evenodd *Calculates the even-odd consistency score*

Description

Takes a matrix of item responses and a vector of integers representing the length each factor. The even-odd consistency score is then computed as the within-person correlation between the even and odd subscales over all the factors. Ported from the **careless** package (version 1.2.2).

Usage

```
evenodd(x, factors, diag = FALSE)
```

Arguments

x	a matrix or data frame of data (e.g. survey responses)
factors	a vector of integers specifying the length of each factor in the dataset
diag	optionally returns a column with the number of available (i.e., non-missing) even/odd pairs per observation. Useful for datasets with many missing values.

Value

If diag = FALSE, a numeric vector of even-odd consistency scores. If diag = TRUE, a data frame containing eo_cor and eo_missing.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Johnson, J. A. (2005). Ascertaining the validity of individual protocols from web-based personality inventories. *Journal of Research in Personality, 39*, 103-129. doi:10.1016/j.jrp.2004.09.009

Examples

```
careless_eo <- evenodd(careless_dataset, rep(5,10))
careless_eodiag <- evenodd(careless_dataset, rep(5,10), diag = TRUE)
```

`filter_respondents` *Filter Respondents Based on Satisficing Flags*

Description

Filters respondent data based on flag indicators using three-valued logic aggregation strategies (`"any"`, `"all"`, or `"at_least"`), requiring explicit respondent ID matching and producing an audit log.

Usage

```
filter_respondents(
  data,
  flags,
  strategy = c("any", "all", "at_least"),
  n = NULL,
  id = NULL,
  na_action = c("error", "keep", "exclude")
)
```

Arguments

`data`	A data frame of survey responses.
`flags`	A data frame containing an ID column and one or more logical flag columns. Flag column names must not conflict with reserved audit column names (`"row_index"`, `"n_flagged"`, `"n_missing_flags"`, `"decision"`, `"strategy"`, `"n_required"`, `"na_action"`, `"excluded"`).
`strategy`	Character string specifying the aggregation strategy: `"any"`, `"all"`, or `"at_least"`.
`n`	An integer specifying the minimum number of flags required for exclusion when `strategy = "at_least"`. Must be `NULL` for other strategies.
`id`	Optional character string specifying the ID column name present in both `data` and `flags`. If `NULL` (default), an existing column named `"id"` present in both data frames is used automatically. The ID column name must not conflict with reserved audit column names (`"row_index"`, `"n_flagged"`, `"n_missing_flags"`, `"decision"`, `"strategy"`, `"n_required"`, `"na_action"`, `"excluded"`).
`na_action`	Character string specifying how to handle indeterminate (`NA`) exclusion decisions: `"error"` (default), `"keep"`, or `"exclude"`.

Details

filter_respondents requires an explicit respondent identifier to ensure safe, unambiguous alignment between survey response data and quality flags. Row-order joining without ID alignment is strictly prohibited.

ID values in both data and flags must be non-missing, unique one-dimensional atomic vectors of identical type and class, with exactly matching ID sets. Flags are aligned to the row order of data by ID.

Respondent exclusion is evaluated using three-valued logic (Kleene logic) across all flag columns before applying na_action:

- "any": Exclude if any flag is TRUE; retain if all flags are FALSE; evaluate to NA if all non-missing flags are FALSE and at least one flag is NA.
- "all": Retain if any flag is FALSE; exclude if all flags are TRUE; evaluate to NA if all non-missing flags are TRUE and at least one flag is NA.
- "at_least": Exclude if the number of TRUE flags is $\geq n$; retain if the sum of TRUE and NA flags is $< n$; evaluate to NA otherwise. Requires integer n between 1 and the number of flag columns.

The na_action parameter determines the resolution for respondents whose aggregated exclusion decision is NA:

- "error" (default): Throws an error if any respondent has an indeterminate (NA) exclusion decision.
- "keep": Treats indeterminate decisions as FALSE (retained).
- "exclude": Treats indeterminate decisions as TRUE (excluded).

The audit table generated in $audit contains fixed metadata columns: row_index, n_flagged, n_missing_flags, decision, strategy, n_required, na_action, and excluded. To prevent unintended overwriting of audit information, neither the selected ID column name nor any flag column name in flags may match any of these reserved column names.

Methodological Note: Respondent filtering decisions can materially impact study conclusions. Researchers should report filtering criteria transparently and conduct sensitivity analyses comparing results with and without flagged respondents.

Value

A named list with three elements:

- data: Data frame of retained respondents, preserving original columns, column order, and row values.
- excluded: Data frame of excluded respondents, preserving original columns, column order, and row values.
- audit: Data frame containing audit records for all respondents in the order of data, with fixed columns: ID, row_index, each flag column, n_flagged, n_missing_flags, decision, strategy, n_required, na_action, and excluded.

Examples

```
survey <- data.frame(id = 1:4, q1 = c(1, 2, 3, 4), q2 = c(5, 4, 3, 2))
flags <- data.frame(
  id = 1:4,
  flag1 = c(FALSE, TRUE, FALSE, NA),
```

```
  flag2 = c(FALSE, FALSE, TRUE, FALSE)
)
filter_respondents(survey, flags, strategy = "any", na_action = "keep")
```

`flag_respondents` *Flag Respondents Based on Logical Criteria*

Description

Evaluates one or more user-defined logical conditions within the context of survey quality indicators, returning a data frame of logical flags.

Usage

```
flag_respondents(indicators, ..., id = NULL)
```

Arguments

`indicators`	A data frame containing respondent-level survey indicators. Column names must be unique.
`...`	One or more logical expressions to evaluate within `indicators`. At least one expression must be provided.
`id`	Optional character string specifying the column name in `indicators` that uniquely identifies respondents. If `NULL` (default), an existing column named `"id"` is retained automatically if present.

Details

Each expression in `...` is evaluated using data-masking within `indicators`, falling back to the calling environment. Expressions must evaluate strictly to a logical vector of length equal to `nrow(indicators)`. Length-1 recycling and implicit type coercion (such as numeric truthy evaluation) are not permitted. Missing values (`NA`) in condition results are preserved.

Resulting flag columns are named after the argument names in `...` if provided, or derived from the deparsed expressions. Column names are preserved without automatic syntactic conversion (`check.names = FALSE`). Duplicate flag names, empty names, or collisions with the retained ID column name result in an error.

If `id = NULL` and a column named `"id"` exists in `indicators`, it is automatically retained at the beginning of the returned data frame. If `id` is explicitly specified, that column is retained at the beginning. Row order of `indicators` is always preserved.

Methodological Note: Quality flags indicate respondent patterns that meet researcher-specified screening rules; they do not definitively establish careless or invalid responding. Screening thresholds should be justified and supported by sensitivity analysis.

Value

A data frame containing the retained ID column (if present or specified) and one or more logical flag columns.

Examples

```
df <- data.frame(
  id = 1:4,
  longstring = c(12, 5, 15, 3),
  inconsistency = c(0.8, 0.2, 0.9, 0.1)
)
flag_respondents(
  df,
  flag_long = longstring >= 10,
  flag_inc = inconsistency >= 0.75
)
```

`irv` *Calculates the intra-individual response variability (IRV)*

Description

The IRV is the "standard deviation of responses across a set of consecutive item responses for an individual" (Dunn, Heggestad, Shanock, & Theilgard, 2018, p. 108). By default, the IRV is calculated across all columns of the input data. Additionally it can be applied to different subsets of the data. This can detect degraded response quality which occurs only in a certain section of the questionnaire (usually the end). With `split = FALSE`, IRV is the standard deviation across the selected response values and is invariant to their order. With `split = TRUE`, consecutive column subsets are formed in their current order, so reordering columns can change subset membership and subset IRV values. Unlike `longstring`, `irv()` does not measure runs of consecutive identical responses. Whereas Dunn et al. (2018) propose to mark persons with *low* IRV scores as outliers - reflecting straightlining responses, Marjanovic et al. (2015) propose to mark persons with *high* IRV scores - reflecting highly random responses (see References). Ported from the **careless** package (version 1.2.2).

Usage

```
irv(x, na.rm = TRUE, split = FALSE, num.split = 3)
```

Arguments

`x`	a matrix or data frame of data (e.g. survey responses)
`na.rm`	logical indicating whether to calculate the IRV for a person with missing values.
`split`	logical indicating whether to additionally calculate the IRV on subsets of columns (of equal length).
`num.split`	the number of subsets the data is to be split in.

Value

If `split = FALSE`, a numeric vector of IRV values. If `split = TRUE`, a data frame containing `irvTotal` and subset columns (`irv1`, `irv2`, etc.).

Author(s)

Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Dunn, A. M., Heggestad, E. D., Shanock, L. R., & Theilgard, N. (2018). Intra-individual Response Variability as an Indicator of Insufficient Effort Responding: Comparison to Other Indicators and Relationships with Individual Differences. *Journal of Business and Psychology, 33(1)*, 105-121. doi:10.1007/s1086901694790

Marjanovic, Z., Holden, R., Struthers, W., Cribbie, R., & Greenglass, E. (2015). The inter-item standard deviation (ISD): An index that discriminates between conscientious and random responders. *Personality and Individual Differences*, 84, 79-83. doi:10.1016/j.paid.2014.08.021

Examples

```
# calculate the irv over all items
irv_total <- irv(careless_dataset)

# calculate the irv over all items + calculate the irv for each quarter of the questionnaire
irv_split <- irv(careless_dataset, split = TRUE, num.split = 4)
boxplot(irv_split$irv4) #produce a boxplot of the IRV for the fourth quarter
```

`longstring` *Identifies the longest string of identical consecutive responses for each observation*

Description

Takes a matrix of item responses and, beginning with the second column (i.e., second item) compares each column with the previous one to check for matching responses. For each observation, the length of the maximum uninterrupted string of identical responses is returned. Additionally, can return the average length of uninterrupted string of identical responses. Ported from the **careless** package (version 1.2.2).

Usage

```
longstring(x, avg = FALSE)
```

Arguments

`x`	a matrix or data frame of data (e.g. item responses)
`avg`	logical indicating whether to additionally return the average length of identical consecutive responses

Value

If `avg = FALSE`, a vector of lengths of the longest uninterrupted string of identical responses. If `avg = TRUE`, a data frame containing `longstr` and `avgstr`.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Johnson, J. A. (2005). Ascertaining the validity of individual protocols from web-based personality inventories. *Journal of Research in Personality, 39*, 103-129. doi:10.1016/j.jrp.2004.09.009

Examples

```
careless_long <- longstring(careless_dataset, avg = FALSE)
careless_avg <- longstring(careless_dataset, avg = TRUE)
boxplot(careless_avg$longstr) #produce a boxplot of the longstring index
boxplot(careless_avg$avgstr)
```

`mahad` *Find and graph Mahalanobis Distance (D) and flag potential outliers.*

Description

Takes a matrix of item responses and computes Mahalanobis D. Can additionally return a vector of binary outlier flags. Mahalanobis distance is calculated using the function `psych::outlier` of the **psych** package, an implementation which supports missing values. Ported from the **careless** package (version 1.2.2).

Usage

```
mahad(x, plot = TRUE, flag = FALSE, confidence = 0.99, na.rm = TRUE)
```

Arguments

`x`	a matrix or data frame of data
`plot`	Plot the resulting QQ graph
`flag`	Flag potential outliers using the confidence level specified in parameter `confidence`
`confidence`	The desired confidence level of the result
`na.rm`	Should missing data be deleted

Value

If `flag = FALSE`, a numeric vector of squared Mahalanobis distances (`d_sq`). If `flag = TRUE`, a data frame containing `d_sq` and boolean `flagged`.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods, 17(3)*, 437-455. doi:10.1037/a0028085

See Also

`psych::outlier` on which this function is based.

Examples

```
mahad_raw <- mahad(careless_dataset) #only the distances themselves
mahad_flags <- mahad(careless_dataset, flag = TRUE) #additionally flag outliers
mahad_flags <- mahad(careless_dataset, flag = TRUE, confidence = 0.999) #Apply a strict criterion
```

---

matching *Identify Rows Matching a Logical Condition*

Description

Evaluates a logical condition within the context of a data frame and adds an integer column indicating which rows satisfy the condition.

Usage

```
matching(data, condition, result_col = "match")
```

Arguments

data	A data frame.
condition	An expression to evaluate within data. Must evaluate to a logical vector whose length is exactly nrow(data).
result_col	A non-empty character string specifying the name of the result column. Defaults to "match". Must not conflict with existing columns in data.

Value

The input data frame with an additional integer column indicating whether each row satisfies the condition:

- 1L: The condition evaluated to TRUE.
- 0L: The condition evaluated to FALSE.
- NA_integer_: The condition evaluated to NA.

Examples

```
df <- data.frame(ID = c(1, 2, 3, 4), Value = c(10, 20, 30, 40))
matching(df, Value > 20)
```

`maxIdenticalRating` *Maximum Identical Rating*

Description

Calculates the proportion of answers taken up by the most common response by the respondent to a set of questions (Kim et al., 2019).

Usage

```
maxIdenticalRating(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

`x`	A data frame or tibble containing survey responses.
`varnames`	A vector of variable names or column indices. If `NULL` (default), all columns of `x` are used.
`verbose`	Logical; if `TRUE` (default), outputs informational messages such as how missing data are handled.
`clean_data`	Logical; should non-numeric columns be converted to numeric? Defaults to `TRUE`.
`round_digits`	A numeric value specifying digits for rounding (kept for API compatibility).

Value

A numeric vector of the same length as `nrow(x)` indicating the proportion of answers that take the most common value.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
maxIdenticalRating(survey, varnames = c("item1", "item4"), verbose = FALSE)
```

`meanRootPairs` *Mean Root Pairs*

Description

Computes the sample-normalized Mean Root of Pairs (MRP) straightlining measure for each respondent across a battery of questions (Kim et al., 2019).

Usage

```
meanRootPairs(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

`x`	A data frame or tibble containing survey responses.
`varnames`	A vector of variable names or column indices. If `NULL` (default), all columns of `x` are used.
`verbose`	Logical; if `TRUE` (default), outputs informational messages.
`clean_data`	Logical; should non-numeric columns be converted to numeric? Defaults to `TRUE`.
`round_digits`	A numeric value specifying digits for rounding (kept for API compatibility).

Details

`meanRootPairs()` strictly implements the mathematical definition from Kim et al. (2019):

$$T_i = \frac{2}{n_i(n_i-1)} \sum_{j<k, j,k \in A_i} \sqrt{|x_{ij} - x_{ik}|}$$

for respondents with at least 2 valid responses, followed by sample-level normalization:

$$\text{MRP}_i = \frac{\max(T) - T_i}{\max(T) - \min(T)}$$

If fewer than 2 respondents have valid T_i values, or if $\max(T) == \min(T)$, `NA_real_` is returned for all rows.

Value

A numeric vector of length `nrow(x)` containing the normalized MRP scores, where higher values represent relatively greater straightlining (lower differentiation).

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(1, 1, 3, 1),
  item2 = c(1, 2, 4, 1),
  item3 = c(1, 3, 5, 2)
)
meanRootPairs(survey, varnames = c("item1", "item2", "item3"), verbose = FALSE)
```

---

mir *Maximum Identical Rating (MIR)*

Description

Computes the Maximum Identical Rating (MIR) straightlining measure across specified columns for each row in a data frame, as described by Kim et al. (2019).

Usage

```
mir(data, columns, result_column_name = "mir")
```

Arguments

`data`	A data frame.
`columns`	A character vector containing at least 2 unique column names existing in `data`. All specified columns must be numeric (integer/double) or all must be factors.
`result_column_name`	A non-empty character string of length 1 specifying the name of the result column to append. Defaults to `"mir"`. Must not already exist in `data`.

Details

`mir()` is a backward-compatibility wrapper around `maxIdenticalRating`. It appends a result column to the input data frame, computing the proportion of answers taking the most common response value across the specified columns.

Value

The original data frame `data` with an additional numeric column named by `result_column_name` containing the MIR scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
df <- data.frame(
  q1 = c(1, 1, 3),
  q2 = c(1, 2, 3),
  q3 = c(1, 3, 3)
)
mir(df, c("q1", "q2", "q3"))
```

`mrp` *Mean Root of Pairs (MRP)*

Description

Computes the Mean Root of Pairs (MRP) straightlining measure across specified columns for each row in a data frame, as described by Kim et al. (2019).

Usage

```
mrp(data, columns, result_column_name = "mrp")
```

Arguments

`data`	A data frame.
`columns`	A character vector containing at least 2 unique column names existing in `data`. Columns must be numeric (integer or double). Factor, character, or logical columns are not permitted.
`result_column_name`	A non-empty character string of length 1 specifying the name of the result column to append. Defaults to `"mrp"`. Must not already exist in `data`.

Details

`mrp()` is a backward-compatibility wrapper around `meanRootPairs`. It appends a result column to the input data frame, computing the sample-normalized Mean Root of Pairs across the specified numeric columns.

Value

The original data frame `data` with an additional numeric column named by `result_column_name` containing the normalized MRP scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
df <- data.frame(
  q1 = c(1, 1, 3, 1),
  q2 = c(1, 2, 4, 1),
  q3 = c(1, 3, 5, 2)
)
mrp(df, c("q1", "q2", "q3"))
```

`nondiff` *Simple Nondifferentiation Indicator*

Description

Evaluates whether a respondent exhibits complete nondifferentiation across a battery of questions (Kim et al., 2019).

Usage

```
nondiff(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

`x`	A data frame or tibble containing survey responses.
`varnames`	A vector of variable names or column indices. If `NULL` (default), all columns of `x` are used.
`verbose`	Logical; if `TRUE` (default), prints the proportion of nondifferentiation.
`clean_data`	Logical; should non-numeric columns be converted to numeric? Defaults to `TRUE`.
`round_digits`	A numeric value specifying digits for rounding printed output.

Details

In accordance with straightliner conventions, `nondiff()` evaluates whether `length(unique(row)) == 1L`. If a row consists entirely of missing values (`NA`), `unique()` has length 1, evaluating to `TRUE`. If a row has both observed values and `NA`, `unique()` has length greater than 1, evaluating to `FALSE`.

Note that this differs from `stline`, which returns an integer flag (`1L`, `0L`, or `NA_integer_`) and evaluates whether all non-missing values are identical when at least 2 non-missing responses are present.

Value

A logical vector indicating whether each respondent gave identical answers to all specified questions (`TRUE`) or not (`FALSE`).

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
nondiff(survey, varnames = c("item1", "item4"), verbose = FALSE)
```

---

print.straightliner *Print Method for straightliner Objects*

Description

Print Method for straightliner Objects

Usage

```
## S3 method for class 'straightliner'
print(x, digits = 2, ...)
```

Arguments

x	An object of class straightliner.
digits	Number of decimal places for rounding output.
...	Additional arguments passed to print.

---

psychant *Computes the psychometric antonym score*

Description

A convenient wrapper that calls psychsyn with argument anto = TRUE to compute the psychometric antonym score. Ported from the **careless** package (version 1.2.2).

Usage

```
psychant(x, critval = -0.6, diag = FALSE)
```

Arguments

x	is a matrix or data frame of item responses
critval	is the minimum magnitude of the correlation between two items in order for them to be considered psychometric synonyms. Defaults to -.60
diag	additionally return the number of item pairs available for each subject. Useful if dataset contains many missing values.

Value

If diag = FALSE, a numeric vector of psychometric antonym scores. If diag = TRUE, a data frame containing numPairs and cor.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

See Also

psychsyn for the main function, psychsyn_critval for a helper that allows to set an adequate critical value for the size of the correlation.

Examples

```
antonyms <- psychant(careless_dataset2, .50)
antonyms <- psychant(careless_dataset2, .50, diag = TRUE)
```

---

psychsyn *Computes the psychometric synonym/antonym score*

Description

Takes a matrix of item responses and identifies item pairs that are highly correlated within the overall dataset. What defines "highly correlated" is set by the critical value (e.g., r > .60). Each respondents' psychometric synonym score is then computed as the within-person correlation between the identified item-pairs. Alternatively computes the psychometric antonym score which is a variant that uses item pairs that are highly *negatively* correlated. Ported from the **careless** package (version 1.2.2).

Usage

```
psychsyn(x, critval = 0.6, anto = FALSE, diag = FALSE, resample_na = TRUE)
```

Arguments

x	is a matrix or data frame of item responses
critval	is the minimum magnitude of the correlation between two items in order for them to be considered psychometric synonyms. Defaults to .60
anto	determines whether psychometric antonyms are returned instead of psychometric synonyms. Defaults to FALSE

diag	additionally return the number of item pairs available for each observation. Useful if dataset contains many missing values.
resample_na	if psychsyn returns NA for a respondent resample to attempt getting a non-NA result.

Value

If diag = FALSE, a numeric vector of psychometric synonym scores (correlations). If diag = TRUE, a data frame containing numPairs and cor.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods, 17(3)*, 437-455. doi:10.1037/a0028085

See Also

psychant for a more concise way to calculate the psychometric antonym score, psychsyn_critval for a helper that allows to set an adequate critical value for the size of the correlation.

Examples

```
synonyms <- psychsyn(careless_dataset, .60)
antonyms <- psychsyn(careless_dataset2, .50, anto = TRUE)
antonyms <- psychant(careless_dataset2, .50)

#with diagnostics
synonyms <- psychsyn(careless_dataset, .60, diag = TRUE)
antonyms <- psychant(careless_dataset2, .50, diag = TRUE)
```

---

psychsyn_critval *Compute the correlations between all possible item pairs and order them by the magnitude of the correlation*

Description

A function intended to help finding adequate critical values for psychsyn and psychant. Takes a matrix of item responses and returns a data frame giving the correlations of all item pairs ordered by the magnitude of the correlation. Ported from the **careless** package (version 1.2.2).

Usage

```
psychsyn_critval(x, anto = FALSE)
```

Arguments

x	a matrix or data frame of item responses.
anto	ordered by the largest positive correlation, or, if anto = TRUE, the largest negative correlation.

Value

A data frame with columns `var1`, `var2`, and `cor`, sorted by correlation magnitude.

Author(s)

Francisco Wilhelm <franciscowilhelm@gmail.com>

See Also

after determining an adequate critical value, continue with `psychsyn` and/or `psychant`

Examples

```
psychsyn_cor <- psychsyn_critval(careless_dataset)
psychsyn_cor <- psychsyn_critval(careless_dataset, anto = TRUE)
```

---

question_sd *Standard Deviation across Question Battery*

Description

Computes the intra-individual standard deviation across a set of questions for each respondent (Kim et al., 2019).

Usage

```
question_sd(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

`x`	A data frame or tibble containing survey responses.
`varnames`	A vector of variable names or column indices. If `NULL` (default), all columns of `x` are used.
`verbose`	Logical; if `TRUE` (default), outputs informational messages regarding missing data handling.
`clean_data`	Logical; should non-numeric columns be converted to numeric? Defaults to `TRUE`.
`round_digits`	A numeric value specifying digits for rounding (kept for API compatibility).

Value

A numeric vector of standard deviations for each respondent across the specified questions.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
question_sd(survey, varnames = c("item1", "item3", "item4"), verbose = FALSE)
```

---

rawnn *Print Column Names with Indices*

Description

Prints the column names of a data frame along with their respective 1-based indices, formatted as `'index: column_name'`.

Usage

```
rawnn(dataframe)
```

Arguments

`dataframe`	A data frame whose column names are to be printed.

Value

`invisible(NULL)`. Called for its side effect of printing.

Examples

```
rawnn(mtcars)
```

---

scale_point_variation *Scale Point Variation*

Description

Computes the Scale Point Variation measure capturing response diversity across questions for each respondent (Kim et al., 2019).

Usage

```
scale_point_variation(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

x	A data frame or tibble containing survey responses.
varnames	A vector of variable names or column indices. If NULL (default), all columns of x are used.
verbose	Logical; if TRUE (default), outputs informational messages, including the battery-level statistic and handling of missing data.
clean_data	Logical; should non-numeric columns be converted to numeric? Defaults to TRUE.
round_digits	A numeric value specifying digits for rounding informational messages.

Value

A numeric vector of length nrow(x) with Scale Point Variation scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
scale_point_variation(survey, varnames = c("item1", "item4"), verbose = FALSE)
```

---

semantic_inconsistency

Semantic Inconsistency across Item Pairs

Description

Evaluates semantic response inconsistency across user-specified item pairs with expected directional relationships ("same" or "opposite"). Inconsistency scores are normalized to the theoretical scale range $[0, 1]$, where 0 represents complete consistency and 1 represents maximal inconsistency.

Usage

```
semantic_inconsistency(
  data,
  pairs,
  scale_min,
  scale_max,
  threshold = 0.75,
  min_valid_pairs = 1L,
  details = FALSE
)
```

Arguments

`data`	A data frame or matrix where rows represent respondents and columns represent survey items.
`pairs`	A data frame with at least columns `item1`, `item2`, and `relation`. Each row defines an item pair. `relation` must be either `"same"` or `"opposite"` (case-sensitive). Self-pairs and duplicate pairs (regardless of item order) are not permitted.
`scale_min`	A finite numeric scalar indicating the theoretical minimum of the response scale.
`scale_max`	A finite numeric scalar indicating the theoretical maximum of the response scale (`scale_min` < `scale_max`).
`threshold`	A finite numeric scalar in $[0, 1]$ used to classify pairs as highly inconsistent (inconsistency $\geq$ `threshold`). Defaults to `0.75` as an illustrative default, not a validated cutoff.
`min_valid_pairs`	An integer scalar ≥ 1 specifying the minimum number of valid pairs required to compute respondent-level summary scores. Defaults to `1L`.
`details`	Logical scalar; if `TRUE`, returns a list containing respondent-level summaries and detailed pair-level evaluations. Defaults to `FALSE`.

Details

For a scale bounded by $[L, U]$ with theoretical range $R = U - L$, the inconsistency between responses x_i and x_j for a valid pair is:

- **Same direction** (`"same"`):

$$\frac{|x_i - x_j|}{U - L}$$

- **Opposite direction** (`"opposite"`):

$$\frac{|(x_i - L) - (U - x_j)|}{U - L}$$

Both formulas are symmetric with respect to item ordering. While algebraically equivalent to $\frac{|x_i + x_j - (L+U)|}{U-L}$, the centered difference formulation $\frac{|(x_i - L) - (U - x_j)|}{U-L}$ is evaluated directly for numerical stability.

When `details` = `FALSE`, the function returns a data frame with one row per respondent containing the following columns in fixed order:

1. `mean_inconsistency` (double): Mean of inconsistency scores across valid pairs.

2. max_inconsistency (double): Maximum inconsistency score among valid pairs.
3. sum_inconsistency (double): Sum of inconsistency scores across valid pairs.
4. n_valid_pairs (integer): Number of pairs where both items have non-missing responses.
5. n_missing_pairs (integer): Number of pairs where at least one item is missing (NA).
6. n_high_inconsistency (integer): Number of valid pairs with inconsistency $\geq$ threshold.
7. prop_high_inconsistency (double): Proportion of high-inconsistency pairs among valid pairs.

If n_valid_pairs < min_valid_pairs, the score and threshold columns (1–3, 6–7) are set to NA of their respective types (NA_real_ or NA_integer_), while n_valid_pairs and n_missing_pairs report the observed counts.

When details = TRUE, the function returns a named list with:

- respondents: The respondent-level data frame described above.
- pairs: A long data frame in row-major order (respondent order of data, and pair order of pairs within each respondent) with columns: respondent_row (integer), pair_id (integer), item1 (character), item2 (character), relation (character), and inconsistency (double).

Methodological Note: Semantic inconsistency scores indicate pattern divergence from researcher-defined item relationships; they do not definitively establish careless or insufficient-effort responding. Items may be included in multiple pairs, but resulting scores are correlated and should not be treated as independent evidence.

Value

When details = FALSE, a data frame of respondent-level summaries. When details = TRUE, a named list with elements respondents and pairs.

Examples

```
df <- data.frame(
  q1 = c(5, 5, 1, 3),
  q2 = c(5, 1, 5, 3),
  q3 = c(1, 5, 5, 3)
)
item_pairs <- data.frame(
  item1 = c("q1", "q1"),
  item2 = c("q2", "q3"),
  relation = c("same", "opposite"),
  stringsAsFactors = FALSE
)
semantic_inconsistency(df, item_pairs, scale_min = 1, scale_max = 5)
```

---

spv *Scale Point Variation (SPV)*

Description

Computes the Scale Point Variation (SPV) straightlining measure across specified columns for each row in a data frame, as described by Kim et al. (2019).

Usage

```
spv(data, columns, result_column_name = "spv")
```

Arguments

`data`	A data frame.
`columns`	A character vector containing at least 2 unique column names existing in `data`. All specified columns must be numeric (integer/double) or all must be factors.
`result_column_name`	A non-empty character string of length 1 specifying the name of the result column to append. Defaults to `"spv"`. Must not already exist in `data`.

Details

`spv()` is a backward-compatibility wrapper around `scale_point_variation`. It appends a result column to the input data frame, computing response diversity across categories for the specified columns.

Value

The original data frame `data` with an additional numeric column named by `result_column_name` containing the SPV scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
df <- data.frame(
  q1 = c(1, 1, 3),
  q2 = c(1, 2, 3),
  q3 = c(1, 3, 3)
)
spv(df, c("q1", "q2", "q3"))
```

`stline` *Check for Straight-Line Responses Across Specified Columns*

Description

Evaluates each row across specified columns in a data frame to identify straight-line responses (identical answers across items).

Usage

```
stline(
  data,
  columns,
  result_column_name = "straight_line",
  exclude_values = NULL
)
```

Arguments

`data`	A data frame. Must not contain duplicate column names.
`columns`	A non-empty character vector of unique column names existing in `data`.
`result_column_name`	A non-empty character string specifying the name of the result column. Defaults to `"straight_line"`. Must not conflict with existing columns in `data`.
`exclude_values`	An optional atomic vector of values to exclude from straight-line detection. If identical values match any value in `exclude_values`, the result is 0. Defaults to `NULL`.

Value

The input data frame with an additional integer column indicating whether the row is a straight-line response:

- `1L`: At least two non-missing values are identical and not excluded.
- `0L`: Non-missing values vary across columns, or identical values match `exclude_values`.
- `NA_integer_`: The row has 0 or 1 non-missing value.

Examples

```
df <- data.frame(
  q1 = c(1, 1, 1, 2, NA),
  q2 = c(1, 1, 2, 2, NA),
  q3 = c(1, 2, 3, 2, 1)
)
stline(df, c("q1", "q2", "q3"))
```

`straightlining` *Individual-Level Straightlining Indicators*

Description

Calculates individual-level indicators of straightlining across a specified set of columns in a survey data frame (Kim et al., 2019).

Usage

```
straightlining(
  x,
  varnames = NULL,
  measures = c("mrp", "mir", "qsd", "spv", "nondiff"),
  keep_original = FALSE
)
```

Arguments

x	A data frame or tibble containing survey responses.
varnames	A vector of variable names or column indices to be analyzed. If NULL (default), all columns of x are used.
measures	A character vector of the measures to compute. Valid options are: • "mrp": Mean Root of Pairs (meanRootPairs). • "mir": Maximum Identical Rating (maxIdenticalRating). • "qsd": Battery Standard Deviation (question_sd). • "spv": Scale Point Variation (scale_point_variation). • "nondiff": Simple Nondifferentiation (nondiff). Defaults to c("mrp", "mir", "qsd", "spv", "nondiff").
keep_original	Logical; if TRUE, returns the original data frame with the computed straightlining indicator columns appended on the right. If FALSE (default), returns only the requested straightlining measures as a data frame.

Value

A data frame containing one column per requested measure. If keep_original = TRUE, all original columns in x are also included.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5, 3),
  item2 = c(5, 5, 2),
  item3 = c(5, 5, 1),
  item4 = c(5, 1, 4)
)
straightlining(survey, varnames = paste0("item", 1:4),
               measures = c("mir", "qsd", "spv", "nondiff"))
```

---

straightlining_qset *Aggregate Straightlining Measures for Question Batteries*

Description

Computes aggregate (battery-level) straightlining measures across multiple sets of questions (Kim et al., 2019).

Usage

```
straightlining_qset(
  x,
  variable_list = NULL,
  measures = c("mrp", "mir", "qsd", "spv", "nondiff"),
  round_digits = 2,
  verbose = FALSE
)
```

Arguments

x	A data frame or tibble containing survey responses.
variable_list	A named list where each element is a vector containing the column names or column indices for one battery of questions.
measures	A character vector of the measures to compute. Valid options are: "mrp", "mir", "qsd", "spv", and "nondiff". Defaults to c("mrp", "mir", "qsd", "spv", "nondiff").
round_digits	A numeric value specifying digits for rounding the printed summary table. Defaults to 2.
verbose	Logical; if TRUE, outputs progress messages per battery. Defaults to FALSE.

Value

A data frame of class c("straightliner", "data.frame") containing the battery-level means for each requested measure (and the maximum SPV "spv_max" if "spv" is requested), returned invisibly.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5, 3),
  item2 = c(5, 5, 2),
  item3 = c(5, 5, 1),
  item4 = c(5, 1, 4)
)
batteries <- list(first_two = c("item1", "item2"), second_two = c(3, 4))
straightlining_qset(survey, variable_list = batteries,
                    measures = c("mir", "qsd", "spv", "nondiff"))
```

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